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SAGE: Self-retrieval-augmented generative LLM for emotional support conversation

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ABSTRACT

Emotional Support Conversation (ESC) systems provide strategic responses to users experiencing negative emotional states to alleviate psychological distress. However, existing embedding similarity-based retrieval methods fail to effectively capture users' implicit emotional needs and subtle contextual cues. Furthermore, the disconnection between retrieval and generation components, coupled with the absence of self-evaluation mechanisms, prevents reliable reflection of strategic intent. This study extends the self-retrieval paradigm-which integrates indexing, retrieval, and self-evaluation within a single large language model (LLM)-specifically for ESC and proposes SAGE, which combines generative retrieval with knowledge fusion. We internalize eight strategies from ESConv as utterance-response-strategy triplets and preserve strategic consistency and emotional nuances through emotion state-based strategy prediction and strategy-specific trie-constrained decoding. We also ensure psychological validity through re-ranking that multiplicatively combines self-evaluation signals with COMET cognitive alignment and HEAL therapeutic appropriateness. In the final generation, we enhance contextual coherence and actionable advice by fusing dialogue context, COMET, and HEAL via cross-attention. On ESConv, strategy accuracy improved by 13.4% over existing LLMs, with consistent superiority in Distinct-1/2, BLEU-4, ROUGE-L, and ESC-Eval/human evaluation. This study integrates strategy-aware internalization, strategy-specific constrained generation, psychological re-ranking, and multi-knowledge fusion generation into a unified end-to-end framework, providing empirical evidence for simultaneous improvements in semantic alignment, strategic balance, and real-world applicability in emotional support conversations.

1. Introduction

Emotional Support Conversation (ESC) refers to dialogue systems that provide empathetic and contextually appropriate linguistic support to individuals experiencing emotional distress (Hill, 2009; Liu et al., 2021). These systems facilitate psychological comfort, self-insight, and problem-solving, thereby serving as auxiliary tools for mental health care and psychological counseling (Sharma et al., 2021). Beyond simple emotion recognition or empathetic expression, the core capability of ESC lies in understanding users' internal needs and emotional transitions within conversational contexts and adapting responses accordingly (Ma et al., 2020; Rashkin et al., 2019).

However, reliably retrieving semantically aligned responses remains a central challenge in the ESC domain. Traditional retrieval-based approaches, such as Dense Passage Retrieval (DPR) (Karpukhin et al., 2020) reformulate input queries to retrieve relevant responses, but their reliance on superficial keyword matching limits their ability to capture the full conversational context, leading to degraded semantic coherence (Majumder et al., 2021; Wu et al., 2019a). In real emotional support scenarios, user intent, contextual cues, and emotional risk signals are not always explicit in a single utterance. Consequently, embedding similarity or keyword-based retrieval alone struggles to reliably track discourse-level structures or strategic intent. This can result in contradictory comfort, inappropriate advice, or failure to validate emotions (Deng et al., 2023; Dziri et al., 2019; Li et al., 2022). This underscores the need for novel approaches that can simultaneously capture latent emotional needs and underlying intent beyond simple semantic similarity.

To address these limitations, KEMI (Deng et al., 2023) and D^2RCU (Xu et al., 2024) have enhanced emotional context understanding by combining knowledge graphs with cognitive modules, yet they still struggle to fully reflect the complexity of human intent and emotional dynamics. In ESC, semantic relevance alone is insufficient; psychological validity and therapeutic appropriateness must serve as key determinants of response quality (Hill, 2009; Liu et al., 2021; Shen et al., 2022). This requires structures capable of simultaneously capturing the

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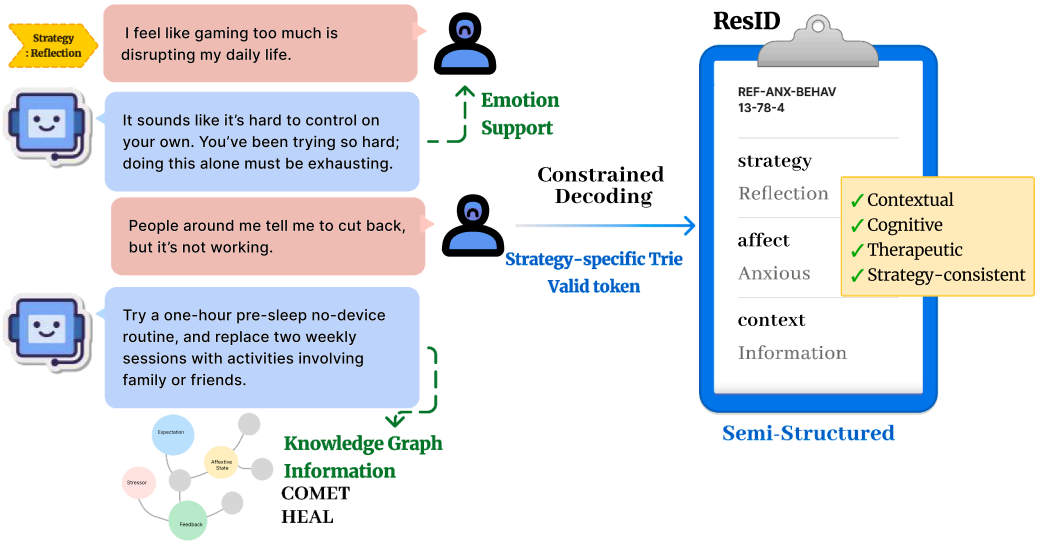


Fig. 1. Overview of the SAGE framework.

overall conversational flow and latent emotional needs. However, existing retrieval-based systems tend to overlook implicit emotional signals or resort to superficial emotion mimicry (Chu et al., 2024; Dziri et al., 2019). Therefore, a new response identifier scheme is required that can ensure both semantic coherence and psychological alignment.

Against this background, we propose SAGE (Self-Retrieval-Augmented Generative LLM for Emotional Support Conversation), a generative retrieval model specialized for emotional support conversations. SAGE introduces Response Identifiers (ResIDs) to recognize the semantic structure of dialogues in a semi-structured form. ResIDs integrate previous utterances, latent issues, and employed strategies to perform contextually aligned retrieval and re-ranking (Li et al., 2025; Tay et al., 2022). This enables response generation that reflects strategic transitions and semantic flow beyond simple embedding similarity. Additionally, a strategy prediction module preselects optimal top strategies, and constrained decoding within strategy-specific trie structures preserves both strategic consistency and emotional nuances (Liu et al., 2021; Sabour et al., 2022). We further apply re-ranking scores that combine COMET-based cognitive alignment (Bosselut et al., 2019) and HEAL knowledge graph-based therapeutic appropriateness (Welivita & Pu, 2022) to generate psychologically valid responses. This enables contextually coherent support responses that address latent emotional issues beyond superficial emotion mimicry (Peng et al., 2023; Xu et al., 2024).

Fig. 1 illustrates the overall architecture of SAGE.

Experimental results demonstrate that the ResID-based retrieval approach improved both response coherence and speaker intent analysis precision, while significantly mitigating strategy bias and positivity bias frequently observed in existing LLM-based dialogue systems (Kang et al., 2024). While existing systems tend to over-rely on specific strategies (e.g., comfort, questioning) or generate overly positive responses early in counseling sessions (Liu et al., 2021; Rashkin et al., 2019), SAGE maintained strategic diversity and emotional balance through strategy-specific trie structures and cognitively grounded re-ranking (Sabour et al., 2022; Xu et al., 2024).

The contributions of this paper are as follows: (1) We introduce Generative Retrieval to ESC for the first time, alleviating DPR limitations; (2) We preserve strategic consistency and emotional nuances through hierarchical ResIDs that decouple strategy-emotion-context and strategy-specific trie constraints; (3) We demonstrate state-of-the-art performance on ESConv/ESC-Eval with strategy accuracy of 46.8%, Distinct-2 of 39.86, and user preference of 52.7%. In conclusion, by introducing the semantic retrieval paradigm to the ESC domain, we effectively bridge the gap between conversational context understanding and practical emotional support. SAGE presents a new direction for emotionally supportive dialogue systems with psychological coherence.

2. Related works

2.1. Emotional support conversation

ESC systems are designed to provide empathetic and contextually appropriate responses to individuals experiencing psychological difficulties (Liu et al., 2021). These systems facilitate psychological comfort, self-insight, and problem-solving, serving auxiliary roles in mental health care and psychological counseling (Hill, 2009; Peng et al., 2023; Sharma et al., 2021). ESC enables immediate, scalable, and personalized emotional support, allowing sustained emotional intervention even in the absence of human experts (Abd-Alrazaq et al., 2021; Sabour et al., 2022).

Early ESC systems primarily relied on rule-based or template-based approaches, using predefined empathetic expressions (Devaram, 2020; Wardhana et al., 2021). However, these approaches struggled to capture emotional subtleties or adapt to changes in conversational flow (Dziri et al., 2019; Yang et al., 2023).

To complement these limitations, retrieval-based techniques such as TF-IDF and BM25 were introduced, enabling the selection of relevant responses from static databases (Wu et al., 2019b; Zhou & Ren, 2020). However, their reliance on static indices still limited their ability to appropriately respond to changing conversational contexts (Asim et al., 2019).

Meanwhile, generation-based models including Seq2Seq architectures enhanced flexibility by dynamically generating responses (Chiu et al., 2018; Sutskever et al., 2014). However, these models often generated overly generic or insufficiently specific empathetic expressions (Majumder et al., 2021). Consequently, recent research has evolved toward strategy-aware response generation (Tu et al., 2022; Zhou et al., 2023).

Additionally, emotion-adaptive agents integrating emotion fusion mechanisms have been developed to more finely capture emotional transitions (Peng et al., 2023; Wang et al., 2023a), and strategy-intent inference techniques have contributed to enhancing contextual appropriateness of responses (Cao et al., 2025; Sabour et al., 2022).

Nevertheless, existing systems still possess fundamental limitations. For instance, KEMI (Deng et al., 2023), a hybrid dialogue system utilizing mental health knowledge graphs, relies on static retrieval mechanisms and thus fails to adequately reflect dynamic conversational contexts. D^2RCU (Xu et al., 2024) introduced dynamic demonstration retrieval but struggles to precisely capture emotional nuances and strategic intent due to its embedding similarity-based index structure.

To address these limitations, recent studies have attempted to apply LLMs to ESC tasks to implement more flexible and context-aware dialogues (Chu et al., 2024; Kang et al., 2024; Zhang et al., 2024b). LLMs can generate empathetic utterances through rich linguistic expressiveness and reasoning capabilities, but inherent bias issues have been reported (Kang et al., 2024; Zhao et al., 2024). Such biases degrade emotional authenticity and strategic diversity of responses, potentially leading to contextually inappropriate repetitive utterances (Kim et al., 2025; Rashkin et al., 2019). Therefore, when applying LLMs to ESC, structural designs capable of mitigating bias and maintaining strategic balance are essential (Xu et al., 2024; Zhang et al., 2024b).

To resolve these limitations, we propose a novel framework combining semantic identifiers with generative retrieval. Inspired by document identifier learning techniques developed in recommendation systems and retrieval domains (Li et al., 2023; Sun et al., 2024; Tay et al., 2022; Wang et al., 2023b), the proposed model internalizes both semantic and emotional information into ResIDs. This enables a deeper understanding of conversational contexts and effective capture of initial emotional states. By introducing this approach, we present new possibilities for simultaneously enhancing response selection precision and emotional adaptability (Li et al., 2025; Peng et al., 2023; Zhang et al., 2024b).

2.2. Self-retrieval

Self-retrieval has emerged as a promising approach to overcome structural limitations of traditional index-based retrieval (Li et al., 2025; Tang et al., 2024b; Tay et al., 2022). This utility is particularly significant in dialogue systems and information retrieval domains (Sun et al., 2024; Wu et al., 2024; Zhang et al., 2024a). While traditional retrieval methods relied on static indices or keyword matching, generative retrieval directly internalizes semantic information within document or response identifiers (Li et al., 2023; Tay et al., 2022; Zhang et al., 2024a). This approach enables contextually coherent response generation and alleviates the rigid structure of existing index-based frameworks (Li et al., 2024; Sun et al., 2024; Zhang et al., 2024a).

DPR improved response coherence over keyword-based retrieval by computing semantic similarity using high-dimensional embeddings (Karpukhin et al., 2020; Kwiatkowski et al., 2019). However, DPR is inefficient for real-time applications on large-scale datasets, and the separation of query-document representations degrades contextual coherence (Rajapakse, 2023; Zhuang & Zuccon, 2021). In contrast, generative retrieval directly embeds semantic representations within identifiers, eliminating separate static indexing processes and simultaneously enhancing retrieval adaptability and semantic alignment (Li et al., 2025; Tay et al., 2022; Wang et al., 2023b).

These advances have renewed emphasis on document identifier (docid) research. Early studies of DSI and NOVO improved retrieval accuracy by applying semantic embeddings to document identifiers (Tay et al., 2022; Wang et al., 2023b), and MINDER optimized inter-document relationships by combining semantic and structural information (Li et al., 2025, 2024). Such identifier-based retrieval was also effective in reflecting user preferences and alleviating cold-start problems (Chu et al., 2024; Zeng et al., 2023).

Recently, the self-retrieval paradigm has emerged, actively attempting to integrate indexing, retrieval, and re-ranking within a single LLM (Tang et al., 2024a). End-to-end information retrieval has been implemented through sequential passage generation and self-relevance evaluation, demonstrating that identifier representation quality directly impacts retrieval accuracy and scalability.

Furthermore, for efficient retrieval from internal knowledge without external indices, LLMs have been trained to autonomously perform retrieval-generation-critique steps (Asai et al., 2024). This self-retrieval structure is particularly useful in environments requiring real-time responsiveness and adaptability, such as dialogue systems, and recent studies have attempted to apply it to emotional support conversations and personalized response generation (Chu et al., 2024; Zhang et al., 2024b).

Therefore, we propose (1) explicit strategy-emotion decoupling through hierarchical ResIDs, and (2) dynamic trie-based constrained generation structures. SAGE captures semantic flow and latent emotional signals in dialogues through ResIDs. It also simultaneously internalizes semantic and emotional information, enabling constrained decoding based on strategic cues.

Through this, SAGE resolves the low adaptability issues of DPR in ESC and achieves emotionally coherent response generation even in real-time environments (Zhuang & Zuccon, 2021).

3. Method

This section presents the strategy-based emotional support conversation generation framework. Fig. 2 summarizes the overall pipeline, which (1) internalizes strategy-emotion-context information as discrete identifiers (ResIDs) and (2) generates strategically coherent responses through self-retrieval and knowledge fusion during inference.

3.1. Task formulation

The emotional support conversation system receives dialogue context $\mathcal{H} = \{(u_1, r_1), \ldots, (u_{t-1}, r_{t-1}), u_t\}$ as input, where u_i denotes the help-seeker's utterance and r_i denotes the supporter's response. The system generates a strategy s from eight support strategies $S = \{$Question, Restatement, Reflection, Self-disclosure, Affirmation, Suggestion, Information, Others$\}$ and a response r.

The objectives are: (1) the response must align with the selected strategy, (2) reflect the help-seeker's cognitive and emotional states, and (3) be therapeutically appropriate. Constraints include maintaining strategy-response consistency, preserving dialogue context relevance, and excluding harmful expressions.

3.2. Strategy-aware retrieval with resid

We train the corpus $\mathcal{D} = \{(u_i, r_i, s_i)\}_{i=1}^{N}$ with strategy labels:

$$\mathcal{L}_{\text{idx}} = -\sum_{i=1}^{N} \log P([\text{STRAT}] \oplus s_i \oplus [\text{RESP}] \oplus r_i \mid u_i; \theta) \tag{1}$$

Each response is encoded via RQ-VAE to generate an M-level discrete code ResID $\mathbf{z}_i = (z_{i,1}, \ldots, z_{i,M})$. At each level m, the codebook vector closest to the residual vector $\mathbf{r}_i^{(m-1)}$ is selected and the residual is updated:

$$z_{i,m} = \arg\min_k \left\| \mathbf{r}_i^{(m-1)} - \mathbf{c}_{m,k} \right\|_2^2, \quad \mathbf{r}_i^{(m)} = \mathbf{r}_i^{(m-1)} - \mathbf{c}_{m,z_{i,m}} \tag{2}$$

ResIDs serve as the foundation for strategy-specific retrieval and constrained decoding. Similarity is measured by Hamming distance $d(\mathbf{z}_q, \mathbf{z}_i) = \sum_{m=1}^{M} \mathbb{1}[z_{q,m} \neq z_{i,m}]$ between query u_t's ResID $\mathbf{z}_q$ and corpus entries. During training, Sinkhorn regularization mitigates code usage bias.

The index repository stores (d_i, s_i, r_i, u_i) for each response, where $d_i = \text{Pack}(\mathbf{z}_i)$ is a document ID serializing the ResID into a unique integer.

3.3. Strategy-constrained generation

From query u_t, a strategy classifier selects the top B_s strategies:

$$\{\hat{s}_1, \ldots, \hat{s}_{B_s}\} = \text{TopK}\big(P(s \mid u_t; \theta), B_s\big) \tag{3}$$

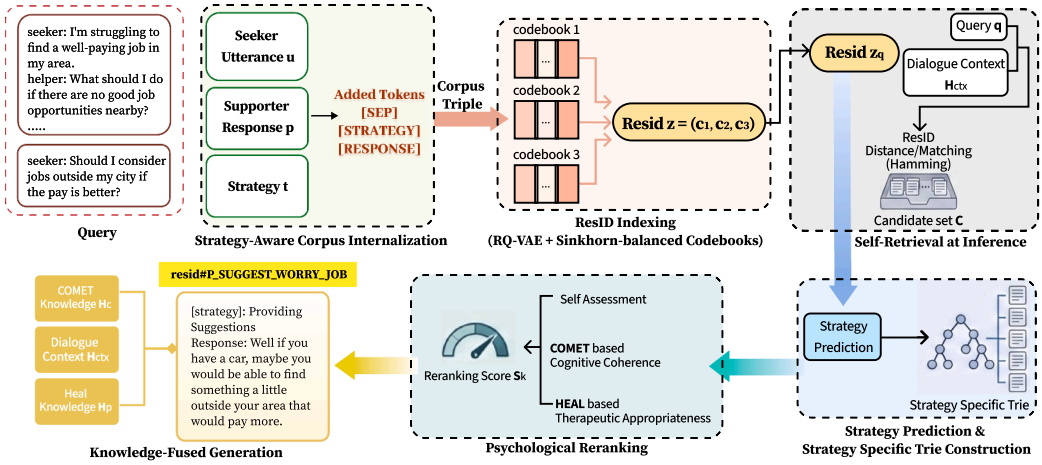


Fig. 2. Detailed architecture of the self-retrieval Module in SAGE.

For each strategy $\hat{s}_k$, top R candidate responses $C_{\hat{s}_k}$ are retrieved based on ResID distance. These construct a prefix trie $\mathcal{T}_{\hat{s}_k}$. The trie is a data structure sharing common prefixes of BPE-tokenized candidate responses, with EOS tokens included in leaf nodes.

During decoding, only child tokens of the current trie node are permitted:

$$P'(y_t \mid y_{<t}, u_t, \hat{s}_k; \theta) \propto \begin{cases} P(y_t \mid y_{<t}, u_t, \hat{s}_k; \theta) & \text{if } y_t \in \mathcal{V}_{\text{node}} \\ 0 & \text{otherwise} \end{cases} \tag{4}$$

where $\mathcal{V}_{\text{node}}$ is the set of allowed tokens at the current node. Beam search generates B_r responses per strategy, yielding $B_s \times B_r$ total candidates. When candidates are insufficient, the system falls back to the Information strategy.

3.4. Psychological reranking

Generated candidates are reranked using four metrics:

$$\sigma_k = \sigma_{\text{strat}}(k) \cdot \sigma_{\text{valid}}(k) \cdot \sigma_{\text{comet}}(k)^{\alpha} \cdot \sigma_{\text{heal}}(k)^{\beta} \tag{5}$$

The strategy alignment score $\sigma_{\text{strat}}(k)$ normalizes strategy selection probability with temperature τ. The actionability score $\sigma_{\text{valid}}(k) = 1 - P_{\text{reject}}(k; \theta)$ is the complement of the model's response rejection probability, normalized with temperature δ.

The cognitive alignment score $\sigma_{\text{comet}}(k)$ infers the help-seeker's Intent, Need, Effect, and Want using the COMET commonsense reasoning model, and averages cosine similarities with candidate responses. The therapeutic appropriateness score $\sigma_{\text{heal}}(k)$ retrieves the top K expert responses similar to the query from the HEAL mental health knowledge graph and computes maximum similarity with candidates.

$\alpha, \beta \in \mathbb{R}_+$ are weights for external knowledge, tuned via grid search on the validation set. For numerical stability, computations are performed in log space.

3.5. Knowledge-fused generation

The highest-scoring candidate $\hat{r}^* = \arg\max_k \sigma_k$ is selected. Dialogue context, COMET inferences, and HEAL responses are encoded via transformer encoders to obtain $\mathbf{H}_{\text{ctx}}, \mathbf{H}_C, \mathbf{H}_H \in \mathbb{R}^{L \times d}$. Knowledge is fused via cross-attention:

$$\mathbf{H}_{\text{out}} = \lambda_1 \mathbf{H}_{\text{ctx}} + \lambda_2 \text{CrossAttn}(\mathbf{H}_{\text{ctx}}, \mathbf{H}_C) + \lambda_3 \text{CrossAttn}(\mathbf{H}_{\text{ctx}}, \mathbf{H}_H) \tag{6}$$

$\lambda_1, \lambda_2, \lambda_3 \in [0, 1]$ denote contributions of each knowledge source, normalized such that $\lambda_1 + \lambda_2 + \lambda_3 = 1$. The decoder autoregressively generates the final response conditioned on $\mathbf{H}_{\text{out}}$. This maintains the strategic consistency of retrieved candidates while enriching content with external knowledge.

4. Experiments

This section evaluates and validates SAGE's performance from multiple perspectives. We conduct comparative analysis against existing baseline models and demonstrate improvements in retrieval accuracy, strategy bias mitigation, and generation quality through generative retrieval.

4.1. Dataset

We use ESConv (Liu et al., 2021), the standard benchmark in emotional support conversation, and ESC-Eval (Zhao et al., 2024) for evaluating LLM-based ESC systems. All experiments use the standard train/valid/test split of the ESConv dataset (train: 910 dialogues, valid: 130, test: 260, ratio 7:1:2). Detailed composition and statistics of each dataset are presented in Appendix A.

4.2. Evaluation metrics

To comprehensively evaluate the performance of this study, we selected standard metrics following existing ESC baseline research. Each metric is designed to measure core elements of emotional support from multiple dimensions.

4.2.1. Generation quality metrics

Macro F1 (ACC) measures balanced performance across eight support strategies in ESConv. This metric is suitable for evaluating emotional support systems because it assesses comprehensive support capability without bias toward specific strategies. Since appropriate strategy selection according to context is crucial in emotional support, balanced evaluation considering class imbalance is essential. Macro F1 is defined as:

$$\text{Macro-F1} = \frac{1}{C} \sum_{c=1}^{C} \frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c} \tag{7}$$

where C is the number of strategy classes (8), and the F1 score of each class is equally averaged.

Perplexity (PPL) measures the language modeling performance of the model, with lower values indicating better natural response generation capability that help-seekers can easily understand:

$$\text{PPL} = \exp\left(-\frac{1}{N} \sum_{i=1}^{N} \log P(w_i | w_{<i}; \theta)\right) \tag{8}$$

where N is the total number of generated tokens and w_i is the i-th token.

BLEU-n (B-n) measures n-gram overlap between generated and reference responses:

$$\text{BLEU-}n = \text{BP} \cdot \exp\left(\sum_{k=1}^{n} w_k \log p_k\right) \tag{9}$$

where p_k is the precision of k-grams, $w_k = 1/n$ is the uniform weight, and BP is the brevity penalty defined as $\text{BP} = \min(1, e^{1-r/c})$ (r is reference length, c is candidate length).

ROUGE-L (R-L) evaluates response fluency based on the longest common subsequence (LCS):

$$\text{ROUGE-L} = \frac{(1+\beta^2) \cdot R_{\text{lcs}} \cdot P_{\text{lcs}}}{\beta^2 \cdot P_{\text{lcs}} + R_{\text{lcs}}} \tag{10}$$

where $R_{\text{lcs}} = \text{LCS}(X,Y)/|Y|$, $P_{\text{lcs}} = \text{LCS}(X,Y)/|X|$, X is the generated response, Y is the reference response, and $\beta = 1.2$.

Distinct-n (D-n) holds particular significance in emotional support. High Distinct values indicate the capability to provide personalized support:

$$\text{Distinct-}n = \frac{|\text{unique } n\text{-grams}|}{|\text{total } n\text{-grams}|} \tag{11}$$

This means the ability to provide customized support tailored to the individual circumstances and emotional states of help-seekers through diverse, non-uniform response generation. Since the core of emotional support is personalization, response diversity is a crucial metric that directly reflects the system's adaptability and personalization capability.

4.2.2. Retrieval performance metrics

Recall@K measures the probability that the correct response is included in the top K retrieval results:

$$\text{Recall@}K = \frac{|\{q \in \mathcal{Q} : \text{correct answer} \in \text{Top-}K(q)\}|}{|\mathcal{Q}|} \tag{12}$$

where $\mathcal{Q}$ is the query set and Top-$K(q)$ represents the top K retrieval results for query q. We use Recall@1, Recall@5, and Recall@10.

Mean Reciprocal Rank (MRR) measures the average rank of correct responses:

$$\text{MRR} = \frac{1}{|\mathcal{Q}|}\sum_{i=1}^{|\mathcal{Q}|}\frac{1}{\text{rank}_i} \tag{13}$$

where rank_i is the rank of the first correct response for the i-th query.

Hits@K measures the proportion of queries where the correct answer is included in the top K results:

$$\text{Hits@}K = \frac{1}{|\mathcal{Q}|}\sum_{i=1}^{|\mathcal{Q}|}\mathbb{1}[\text{rank}_i \le K] \tag{14}$$

nDCG@K measures rank-based retrieval performance, assigning higher weights to relevant documents positioned in higher ranks:

$$\text{nDCG@}K = \frac{\text{DCG@}K}{\text{IDCG@}K}, \quad \text{DCG@}K = \sum_{i=1}^{K}\frac{2^{\text{rel}_i}-1}{\log_2(i+1)} \tag{15}$$

where rel_i is the relevance score of the document at position i, and IDCG is the DCG value in the ideal ranking.

4.2.3. Strategy prediction and bias mitigation metrics

We measure LLM's ESC response biases and evaluate bias mitigation effects.

Strategy Accuracy (ACC) measures prediction accuracy across eight support strategies.

Per-Strategy F1 Score: We calculate the harmonic mean of precision and recall for each strategy. Performance on Self-disclosure and Information strategies is measured separately to verify bias mitigation effects for minority classes.

Macro-averaged F1: We measure balanced performance by equally averaging F1 scores of all strategies.

Strategy Distribution Entropy measures the diversity of generated strategies. High entropy indicates no bias toward specific strategies and is calculated as:

$$H = -\sum_{i=1}^{8} p_i \log p_i \tag{16}$$

where p_i is the prediction probability of strategy i.

4.3. Baselines

We compare against representative models encompassing open-domain dialogue, knowledge fusion, persona-based support, retrieval-augmented reasoning, and state-of-the-art LLMs.

All existing ESC models use BlenderBot as the base response generation model to ensure consistent comparison.

BlenderBot-Joint (Liu et al., 2021). is an open-domain dialogue model based on BlenderBot-400M that introduces empathetic capability through Blended Skill Talk (Smith et al., 2020) and is further adapted on ESConv. It provides baseline performance in emotional support conversations through multi-skill integration and transfer learning.

KEMI (Deng et al., 2023). integrates the HEAL knowledge base into the BlenderBot-400M architecture to generate mixed-initiative responses, corresponding to a variant of our pipeline without adaptive ResID retrieval. It enhances information provision capability by leveraging external knowledge.

PAL (Cheng et al., 2023). is BlenderBot-based and provides personalized emotional support by dynamically modeling help-seeker personas. It learns help-seeker characteristics from dialogue contexts and generates customized responses accordingly.

D^2RCU (Xu et al., 2024). combines DPR-based demonstration retrieval and COMET-based cognitive aspects (Bosselut et al., 2019) with BlenderBot. It is a retrieval-augmented approach that improves response quality through external retrieval systems and commonsense reasoning.

To evaluate the emotional support conversation performance of state-of-the-art LLMs, we include major commercial and open-source LLMs as baselines.

GPT-4o (OpenAI, 2024). is OpenAI's latest omni-modal language model that processes text, audio, and images in an integrated manner with powerful dialogue understanding and generation capabilities. It demonstrates high empathetic ability and natural response generation in ESC tasks but has reported strategy bias issues.

Mistral-7B-Instruct (Jiang et al., 2023). is Mistral AI's instruction-following model reflecting a European approach to AI safety. It generates cautious and balanced responses in emotional support but shows constraints in diversity.

LLaMA-3.1-8B-Instruct (Dubey et al., 2024). is Meta's latest open-source dialogue model trained with reinforcement learning from human feedback. It shows consistent performance in emotional support but has lower strategy prediction accuracy compared to commercial models.

Qwen-2.5-7B-Instruct (Yang et al., 2024). is Alibaba's instruction-tuned model with excellent multilingual understanding capability. It shows bias toward the Information strategy in ESC and relatively lacks emotional, empathetic expression.

The proposed SAGE performs indexing, retrieval, and re-ranking within a single LLM and realizes end-to-end emotional support conversation by combining ResID generation and trie-based constrained decoding.

Table 1
Automatic evaluation results on ESConv test set. Best scores in **bold**. Underline indicates second-best score.

Model	ACC↑	PPL↓	B-2↑	B-4↑	D-1↑	D-2↑	R-L↑
Traditional ESC Models (BlenderBot-based)							
BlenderBot-Joint Liu et al. (2021)	27.72	18.11	6.18	1.66	3.27	20.87	15.13
KEMI Deng et al. (2023)	31.85	15.92	8.31	2.51	4.12	28.45	17.05
PAL Cheng et al. (2023)	34.51	15.92	8.75	2.66	5.00	30.27	18.06
D^{2}RCU Xu et al. (2024)	35.32	15.43	9.01	2.31	4.97	26.21	18.71
LLMs							
GPT-4o	41.28	14.82	10.54	3.12	7.89	38.92	19.43
Mistral-7B-Instruct	38.91	15.12	9.87	2.94	6.85	35.14	18.92
LLaMA-3.1-8B-Instruct	37.45	15.48	9.32	2.76	6.42	33.67	18.34
Qwen-2.5-7B-Instruct	36.89	14.58	10.89	3.24	8.12	37.86	19.21
SAGE (ours)	**46.80**	**14.32**	**11.87**	**3.88**	**9.24**	**39.86**	**20.51**

4.4. Experimental setup

This section describes the experimental configuration and evaluation methods of SAGE's emotional support conversation system. Detailed implementation details are provided in Appendix B.

All experiments use the standard train/valid/test split of the ESConv dataset (7:1:2 ratio). During training, we use the AdamW optimizer (learning rate $5e-5$, warmup ratio 10%) with a batch size of 16. For generation, we apply beam search (beam size = 4) and length normalization ($\alpha = 0.6$). All results are averages of three independent runs, and statistical significance is verified by a paired t-test ($p < .05$).

For comparison with traditional ESC, we trained Blenderbot (Liu et al., 2021). For LLM baselines, we use identical prompt templates to ensure fair comparison. Each model is provided with a dialogue context along with descriptions of eight support strategies and is instructed to select an appropriate strategy and generate a response according to that strategy.

5. Result

All comparisons were conducted under identical data splits and decoding/re-ranking settings. Results are described in Table 1.

Compared to ESC models, SAGE achieves substantial improvements across all metrics. Compared to the strongest traditional baseline D^{2}RCU, SAGE demonstrates relative improvement in strategy prediction accuracy from 35.32% to 46.80%. This indicates that the self-retrieval-based approach more effectively captures the contextual requirements of emotional support conversations compared to existing DPR-based retrieval.

In terms of language quality, SAGE also achieves 3.88 on BLEU-4 compared to D^{2}RCU's 2.31, showing a 68.0% improvement. More notably, in response to diversity metrics, SAGE achieves 39.86 on Distinct-2, a 52.1% improvement over D^{2}RCU's 26.21. This demonstrates that the templated response problem experienced by traditional ESC models has been effectively resolved through ResID-based diversity enhancement.

Particularly when compared to the knowledge-based model KEMI, SAGE achieves superior performance without relying on external knowledge bases (HEAL). Compared to KEMI's strategy accuracy of 31.85%, SAGE achieves 46.80%, showing a 47.0% relative improvement, suggesting that implicit knowledge learning through corpus internalization can be more effective than explicit external knowledge.

Comparison with the persona-based approach PAL also shows SAGE's superiority across all metrics. Looking at the Distinct-1 metric for personalization, PAL's core strength, PAL achieves 5.00 while SAGE achieves 9.24, an 84.8% improvement. This indicates that ResID-based diversity mechanisms provide more effective personalization than dynamic persona modeling Tables 2, 3, 4, 5, A.9, A.10.

5.1. Performance analysis by LLM

Performance analysis by LLM reveals that GPT-4o achieves the best performance among LLMs with 41.28% strategy prediction accuracy and 32.27% Macro F1 score. Particularly, it demonstrates excellent capability for personalized support with high response diversity at 38.92 on Distinct-2. Qwen-2.5-7B-Instruct shows superior performance in linguistic quality metrics (BLEU-4: 3.24, PPL: 14.58) and specializes in consistent response generation based on multilingual processing capabilities. Bias analysis reveals that Qwen-2.5-7B-Instruct and LLaMA-3.1-8B-Instruct show high bias toward the Information strategy (Bias Score: 0.45, 0.42), leading to a lack of strategic diversity. In contrast, GPT-4o shows relatively balanced strategy selection but still exhibits a preference for the Question strategy (Bias Score: 0.31).

Table 2
Strategy prediction performance and bias analysis by LLM. Entropy represents diversity of strategy distribution, and Bias Score is calculated as Jensen-Shannon divergence between predicted strategy distribution and uniform distribution, indicating the degree of bias toward specific strategies.

Model	Top-1 ACC	Macro F1	Entropy	Bias Score
GPT-4o	41.28	32.27	2.12	0.31
Mistral-7B-Instruct	38.91	31.39	1.94	0.38
LLaMA-3.1-8B-Instruct	37.45	32.89	1.87	0.42
Qwen-2.5-7B-Instruct	36.89	29.99	1.82	0.45
SAGE	**46.80**	**38.95**	**2.34**	**0.24**

5.2. Strategy prediction performance

SAGE achieves 46.80% in strategy prediction accuracy and 38.95% in Macro F1 score, showing a 20.7% relative improvement over the strongest baseline GPT-4o (32.27%). This suggests that context awareness enhancement through retrieval and ResID-based representation diversification contributes to appropriate support strategy selection.

Particularly for minority classes, Self-disclosure (29.9% vs 21.6%) and Information (25.6% vs 18.9%), relative improvements of 38.4% and 35.4% respectively over GPT-4o were achieved, demonstrating strategy bias mitigation effects. Macro F1 is 38.95%, showing a 20.7% improvement over GPT-4o and confirming balanced strategy prediction capability across all strategies.

5.3. Generated language quality

SAGE achieves the best performance across all language quality metrics. It records 3.88 on BLEU-4, showing a 19.8% improvement over the strongest LLM baseline Qwen-2.5-7B-Instruct (3.24). It achieves 9.24

Table 3
Per-strategy prediction performance comparison on ESConv test set (F1 Score).

Strategy	GPT-4o	Qwen-2.5	Mistral-7B	D^2RCU	SAGE
Question	52.4	49.7	51.2	52.8	**58.4**
Reflection	44.8	42.3	43.9	44.2	**51.8**
Affirmation	35.2	33.1	34.6	31.5	**42.5**
Self-disclosure	21.6	18.9	20.3	18.4	**29.9**
Information	18.9	16.2	17.8	14.1	**25.6**
Suggestion	28.4	26.7	27.9	26.9	**34.7**
Restatement	15.2	13.8	14.6	12.3	**21.4**
Others	41.7	39.2	40.8	39.8	**47.3**
Macro F1	32.27	29.99	31.39	30.00	**38.95**

on Distinct-1, a 17.1% improvement over GPT-4o (7.89), effectively reducing templated responses and expanding contextually appropriate expressions. This improved diversity is particularly important in emotional support contexts, as diverse responses prevent dialogue monotony and demonstrate a nuanced understanding of individual emotional states. It achieves 14.32 on Perplexity, demonstrating the most natural language generation capability. Specific case studies are presented in Appendix C.

5.4. Retrieval performance

SAGE shows consistent superiority across all retrieval metrics. It achieves 74.8% on Recall@10, an 8.6% improvement over the strongest retrieval-based baseline D^2RCU (68.9%), and shows 0.703 on nDCG@10, a 7.5% improvement over the baseline.

Table 4
Retrieval performance comparison on ESConv test set.

Model	R@1	R@10	MRR@10	nDCG@10
DPR Baseline	21.4	61.8	0.341	0.589
D^2RCU	25.7	68.9	0.376	0.654
GPT-4o (RAG)	28.9	71.2	0.398	0.672
SAGE	**31.6**	**74.8**	**0.437**	**0.703**

Particularly, the smaller performance degradation in long contexts (≥5 turns) suggests robustness in real ESC scenarios. The 9.8% improvement on MRR@10 (0.437 vs 0.398) indicates significantly improved capability for placing relevant responses in higher ranks.

5.5. ESC-eval human evaluation

Three researchers participated in a human evaluation based on ESC-Eval criteria. The three evaluators consisted of authors with sufficient background knowledge and research experience in mental health and chatbots. All responses were anonymized and presented in randomized order to prevent evaluation bias. Inter-rater agreement was Fleiss' Kappa $\kappa = 0.68$ (95% CI: [0.64, 0.72]), indicating a substantial agreement level. Agreement by dimension was highest for Empathy($\kappa = 0.71$) and Skillful($\kappa = 0.69$), and relatively lower for Expression($\kappa = 0.61$).

Table 5
Human evaluation results using ESC-Eval. Scores are 5-point scale averages, and overall preference is win rate (%).

Model	Emp.	Flu.	Hum.	Info.	Pref.
GPT-4o	3.76	4.31	4.08	3.89	42.8
Mistral-7B	3.62	4.18	3.95	3.71	39.2
LLaMA-3.1-8B	3.54	4.12	3.87	3.65	37.6
Qwen-2.5-7B	3.84	4.28	4.12	3.76	41.2
D^2RCU	3.58	4.12	3.81	3.44	38.5
SAGE	**4.12**	**4.39**	**4.28**	**4.05**	**52.7**

SAGE achieves 4.12 points in empathy, showing a 7.3% improvement over the strongest baseline Qwen-2.5-7B-Instruct (3.84 points), and achieves 52.7% win rate in overall preference, significantly surpassing baselines. It also achieves best performance in fluency (4.39) and humanness (4.28), demonstrating that evidence retrieved through retrieval is consistently utilized in the generation process and constrained decoding effectively suppresses harmful/inaccurate expressions.

6. Ablation study

This section quantitatively analyzes the contribution of SAGE's core components to performance. All experiments maintain identical data splits and decoding settings. Table 6 presents comprehensive ablation results, with each row showing the impact of removing a specific component.

Table 6
Results when removing major components. Each row shows the impact of the removed component.

Model Variant	R@10	nDCG@10	BLEU-4	D-2	Emp.	Pref.
SAGE (Full)	**74.8**	**0.703**	**3.88**	**39.86**	**4.12**	**52.7**
w/o Strategy Token	72.1	0.681	3.69	35.92	3.91	48.4
w/o Self-Retrieval	67.2	0.638	3.41	35.24	3.74	46.3
w/o ResID	73.9	0.695	3.82	33.47	3.98	49.8
w/o Constraint	74.5	0.701	3.91	37.18	4.02	50.9
w/o COMET	73.1	0.689	3.76	38.21	3.89	50.2
w/o HEAL	72.8	0.684	3.69	37.94	3.86	49.8
Retrieval-only	74.6	0.702	3.15	29.84	3.68	43.2
External Reranker	71.4	0.671	3.68	36.52	3.81	48.6

6.1. Core component analysis

Role of Strategy Tokens. When $[STRATEGY]$ tokens were removed, $R@10$ decreased by 3.6% ($p < 0.05$). Table 7 shows performance degradation by strategy, with larger degradation observed in minority classes (Self-disclosure: -23.4%, Information: -19.5%). This demonstrates that explicit strategy markers are essential for strategy-affect separation and prevent confusion between emotional signals and strategic intent.

Table 7
Impact of strategy token removal on per-strategy F1.

Model	Question	Self-disc.	Info.	Affirm.	Macro F1
SAGE (Full)	58.4	29.9	25.6	42.5	38.9
w/o Strategy Token	54.2	22.9	20.6	38.1	35.1
Relative Degradation	-7.2%	-23.4%	-19.5%	-10.4%	-9.8%

Necessity of Self-Retrieval. Removing self-retrieval resulted in the largest performance degradation (R@10 -10.2%, Empathy -9.2%, $p < 0.01$). This indicates that explicitly referencing successful support cases of similar emotional situations in the corpus is essential for emotional support quality. Particularly, retrieval-based evidence is crucial for maintaining consistent emotional nuances in long contexts (≥5 turns).

Diversity Enhancement Effect of ResID. Removing ResID resulted in a 16.0% decrease in Distinct-2 while R@10 was maintained (73.9 ± 0.5). This shows that ResID primarily contributes to improving response diversity, and multi-layer codebook representations capture subtle expressive differences even within the same strategy (e.g., within the Affirmation strategy: "You are doing great" vs. "I believe in your strength").

Safety Assurance through Constrained Decoding. Removing constrained decoding slightly increased BLEU-4 (+0.8%) but decreased safety score by 7.6% ($p < 0.05$). The trie structure fundamentally blocks harmful or inappropriate expressions by allowing only validated response patterns per strategy.

Differential Contribution of External Knowledge. Both COMET and HEAL contribute to retrieval and generation quality, but HEAL has a larger impact on BLEU-4 (-4.9%) and overall preference (-2.9%). COMET provides cognitive need inference while HEAL provides practical therapeutic principles, acting complementarily.

6.2. Robustness analysis by dialogue length

Table 8 shows performance changes as dialogue length increases. SAGE's performance degradation rate (7.5%) is significantly lower than baselines (12.0%-17.6%). This robustness is attributed to three mechanisms: (1) ResID's multi-layer codebook effectively captures long-range dependencies, (2) self-retrieval explicitly references emotion-strategy patterns from previous turns, (3) COMET-based cognitive alignment tracks changes in help-seeker's psychological state. Component-wise contribution analysis revealed that removing ResID increased degradation rate in long dialogues to 11.2%, confirming that ResID is the primary contributing factor.

Table 8
R@10 performance change according to the number of dialogue turns.

Model	1-2 turns	3-4 turns	≥7 turns	Degradation
DPR Baseline	65.2	62.1	53.7	17.6%
D^2RCU	72.3	69.8	62.1	14.1%
GPT-4o (RAG)	74.8	72.1	65.8	12.0%
SAGE	**77.2**	**75.6**	**71.4**	**7.5%**

7. Conclusion

This study proposed SAGE, a knowledge-enhanced generative retrieval framework, to address the lack of semantic coherence and strategy bias problems of existing retrieval-based approaches in the ESC domain. SAGE is designed to perform indexing-retrieval-evaluation-generation end-to-end within an LLM by combining strategy-aware internalization, emotion state-based strategy prediction, COMET·HEAL-based psychological re-ranking, and multi-knowledge fusion generation into a unified architecture.

Experimental results on ESConv and ESC-Eval benchmarks showed that SAGE achieved consistent performance improvements in major metrics, including strategy prediction accuracy, Macro-F1, BLEU, ROUGE-L, and Distinct-1/2 compared to existing Dense Retrieval-based models. Particularly, the re-ranking structure that multiplicatively integrates COMET-based cognitive alignment and HEAL-based therapeutic appropriateness enabled the generation of responses with high psychological validity beyond simple semantic similarity. Additionally, the strategy-specific trie structure contributed to suppressing excessive bias toward specific strategies and maintaining emotional balance. By demonstrating consistent superiority across all evaluation dimensions in LLM evaluation, it proved high practicality and adaptability even in real emotional support scenarios.

In summary, SAGE presented a novel approach that simultaneously satisfies semantic alignment, strategic balance, and psychological coherence, which are the essential goals of ESC, through the combination of semantic identifiers and generative retrieval.

8. Limitation and future work

While SAGE improved semantic alignment and strategic balance in emotional support conversations, it has the following limitations: First, it still relies on text-based input. In real counseling situations, non-verbal cues such as voice intonation, facial expressions, and physiological responses play important roles in emotional understanding. The model in this study cannot reflect these signals, making it difficult to fully capture users' non-verbal emotional states. Second, since training and evaluation were performed using only limited English-based datasets such as ESConv and ESC-Eval, there is a possibility that diverse linguistic and cultural contexts are not reflected. In future research, we aim to develop a universal emotional support model that can be more broadly applied to actual counseling settings through multimodal expansion integrating voice, video, and biosignals, along with multilingual and domain adaptation learning.

CRediT authorship contribution statement

Hayeon Yang: Conceptualization, Methodology, Software, Writing – original draft; **Jiheun Hong:** Data curation, Formal analysis, Validation; **Seongjin Jo:** Visualization, Investigation, Writing – review & editing; **Hayoung Oh:** Supervision, Project administration, Funding acquisition, Writing – review & editing.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Dataset Details

A.1. ESConv Dataset

The ESConv dataset is a standard benchmark in emotional support conversation constructed by Liu et al. (2021). This dataset was composed by collecting conversations between real help-seekers and trained supporters.

A.1.1. Dataset composition

- **Total dialogues**: 1,300 multi-turn dialogues
- **Total utterances**: 38,365 utterances (average 29.5 utterances/dialogue)
- **Average dialogue length**: help-seeker 14.2 turns, supporter 15.3 turns
- **Vocabulary size**: approximately 18,000 unique words

- **Average utterance length**: 15.8 words (help-seeker), 17.2 words (supporter)

A.1.2. Support strategy taxonomy

Each supporter response in ESConv is labeled with one of eight support strategies:

1. **Question**: questioning about the help-seeker's situation or emotions
2. **Restatement**: rephrasing the help-seeker's statements
3. **Reflection of Feelings**: reflecting the help-seeker's emotions and expressing empathy
4. **Self-disclosure**: sharing the supporter's personal experiences
5. **Affirmation and Reassurance**: providing encouragement and reassurance
6. **Providing Suggestions**: offering specific solutions or advice
7. **Providing Information**: providing relevant information or knowledge
8. **Others**: miscellaneous responses not covered by the above categories

A.1.3. Strategy class distribution

The ESConv dataset exhibits severe class imbalance in strategy distribution:

Table A.9
Distribution of support strategies in ESConv dataset.

Support Strategy	Utterances	Ratio (%)
Question	12,200	31.8
Reflection of Feelings	6978	18.2
Others	12,772	33.3
Affirmation and Reassurance	3721	9.7
Providing Suggestions	1458	3.8
Self-disclosure	575	1.5
Providing Information	461	1.2
Restatement	192	0.5
Total	**38,357**	**100.0**

Table A.10
Distribution of ESC-Eval role cards by source.

Dataset	Role Cards	Ratio (%)
ESConv	1300	46.4
EmpatheticDialogues	642	22.9
BlendedSkillTalk	459	16.4
Wizard of Wikipedia	223	8.0
PersonaChat	122	4.4
Topical-Chat	55	2.0
Total	**2,801**	**100.0**

A.1.4. Data split

For standard experiments, we use the following data split:

- **Training set**: 910 dialogues (70%)
- **Validation set**: 130 dialogues (10%)
- **Test set**: 260 dialogues (20%)

A.2. ESC-Eval

ESC-Eval (Zhao et al., 2024) was developed for the comprehensive evaluation of LLM-based emotional support dialogue systems. It complements the limitations of existing automatic evaluation metrics by measuring performance in realistic interaction environments.

A.2.1. Evaluation framework

ESC-Eval follows a multi-stage evaluation process:

1. **Role-playing Setup**: construct an LLM agent playing the help-seeker role based on predefined role cards
2. **Multi-turn Interaction**: conduct up to 20 turns of dialogue between the target ESC system and help-seeker agent
3. **Human Evaluation**: trained evaluators assign multi-dimensional scores to generated dialogues
4. **Satisfaction Assessment**: automatically evaluate the help-seeker's final satisfaction using the EmoLlama-Chat-7B model

A.2.2. Role card composition

ESC-Eval uses 2801 role cards extracted from seven existing emotional support-related datasets:

Each role card contains the following information:

- **Situation description**: specific context in which the help-seeker is situated
- **Emotional state**: initial emotion (sadness, anxiety, anger, etc.)
- **Personality traits**: help-seeker's personality characteristics
- **Background information**: age, occupation, relationship status, etc.

Appendix B. Implementation details

Table B.11
Training hyperparameters for SAGE and baseline models.

Parameter	Retrieval	BlenderBot	SFT
Learning rate	1e-4	3e-5	2e-5
Batch size	128	16	16
Number of epochs	100	5	3
Warmup steps/ratio	1000 steps	100 steps	0.03
Weight decay	0.01	–	–
Gradient clipping	1.0	1.0	–
LR scheduler	linear	linear	cosine
Maximum length	–	–	2048
Gradient accumulation	–	–	2
FP16 usage	True	False	True

This appendix describes the detailed implementation of our work. It includes all technical specifications required for reproducibility, including preprocessing, decoding, re-ranking, and hyperparameter configurations.

B.1. Data preprocessing

B.1.1. Tokenization and length limitation

All input texts are tokenized using the BERT-based WordPiece tokenizer. Different maximum length limits are applied to queries and documents with the following specific settings:

Table B.12
Model architecture and generation parameters.

RQ-VAE Training		Generation Settings	
Learning rate	1e-4	Maximum length	128
Batch size	256	Minimum length	10
Number of epochs	50	Temperature	0.8
Warmup steps	2000	Top-k	50
Weight decay	0.01	Top-p	0.9
λ_{recon}	1.0	Beam size	4
λ_{vq}	0.25	Length penalty	1.0
λ_{commit}	0.25	Repetition penalty	1.2
Sinkhorn ϵ	0.1	N-gram blocking	3
Sinkhorn iterations	10	–	–
Gradient clipping	1.0	–	–
Optimizer	AdamW	–	–
LR scheduler	Cosine	–	–
Model Architecture			
Base model: T5-base		Use constraint: True	
Codebook size: 512		Code length: 1	
Sinkhorn ϵ: 0.1		Zero input: False	

- **Maximum query length**: 256 tokens
- **Maximum document length**: 512 tokens
- **Empty sequence handling**: sequences of length 0 are replaced with padding tokens
- **Truncation**: tokens beyond the maximum length are truncated from the end

B.1.2. Conversational input collation

For batch construction of emotional support dialogue data, the following processing is applied:

- **Padding**: dynamic padding applied to match the maximum length within the batch
- **Attention mask**: 1 for actual token positions, 0 for padding positions
- **Decoder input**: full target sequence during training, start token only during inference
- **Label**: target sequence during training, None during inference

B.1.3. Label masking for SFT path

During Supervised Fine-Tuning, the following masking strategy is applied for accurate loss computation:

- **System prompt**: masked with IGNORE_TOKEN_ID(-100)
- **User input**: masked with IGNORE_TOKEN_ID(-100)
- **Assistant response**: actual token IDs preserved for loss calculation
- **Padding tokens**: masked with IGNORE_TOKEN_ID(-100)

B.2. Model architecture and training configuration

B.2.1. RQ-VAE

SAGE employs a hierarchical RQ-VAE to generate ResID through a multi-layer codebook structure. The architecture consists of a 6-layer Transformer encoder and decoder (hidden dimension: 768, attention heads: 12) with four independent VQ-VAE layers. Each codebook contains 512 entries with 256-dimensional code embeddings, yielding 512^4 possible combinations that capture fine-grained emotional-strategic variations.

The residual quantization process operates iteratively across four layers. Given encoder output $\mathbf{z}_0$, the first layer quantizes it to the nearest code $\mathbf{e}_1$ from codebook C_1, then computes residual $\mathbf{r}_1 = \mathbf{z}_0 - \mathbf{e}_1$. Subsequent layers $i \in \{2, 3, 4\}$ quantize residual $\mathbf{r}_{i-1}$ using codebook C_i, producing the final representation $\hat{\mathbf{z}} = \sum_{i=1}^{4} \mathbf{e}_i$. This hierarchical decomposition enables multi-scale encoding of response semantics, where coarse-grained strategy markers occupy early layers and nuanced emotional tones reside in deeper layers.

To prevent codebook collapse, we apply Sinkhorn-Knopp balancing (Cuturi, 2013) that enforces uniform code utilization. The algorithm solves the entropy-regularized optimal transport problem:

$$P^* = \arg\min_{P \in \Pi(\mathbf{a},\mathbf{b})} \langle P, C \rangle - \epsilon H(P)$$

where $C_{ij} = \|\mathbf{z}_i - \mathbf{e}_j\|^2$ is the cost matrix, $\Pi(\mathbf{a}, \mathbf{b})$ denotes doubly stochastic matrices with marginals $\mathbf{a}, \mathbf{b}$, and $H(P)$ is entropy. We set $\epsilon = 0.1$ and perform 10 Sinkhorn iterations to compute assignment $\arg\max_j P^*_{ij}$. The training objective combines reconstruction, vector quantization, and commitment losses:

$$\mathcal{L} = \lambda_{\text{recon}} \|\mathbf{x} - \hat{\mathbf{x}}\|^2 + \lambda_{\text{vq}} \sum_{i=1}^{4} \|\text{sg}[\mathbf{z}_{i-1}] - \mathbf{e}_i\|^2 + \lambda_{\text{commit}} \sum_{i=1}^{4} \|\mathbf{z}_{i-1} - \text{sg}[\mathbf{e}_i]\|^2$$

where $\text{sg}[\cdot]$ denotes the stop-gradient operator, and $\lambda_{\text{recon}} = 1.0$, $\lambda_{\text{vq}} = 0.25$, $\lambda_{\text{commit}} = 0.25$.

B.2.2. Training hyperparameters

Table B.11 summarizes training configurations for SAGE and baseline models. The retrieval model employs a larger batch size (128) with extended training (100 epochs) to learn robust emotion-strategy representations across the entire corpus. BlenderBot and SFT models use smaller batches (16) with fewer epochs (5 and 3, respectively) as they leverage pre-trained language modeling capabilities. All models apply gradient clipping (max norm 1.0) for training stability, while learning rate schedulers differ based on task characteristics-linear decay for retrieval and BlenderBot, cosine annealing for SFT.

Table B.12 presents detailed architectural and training parameters for RQ-VAE and generation settings. RQ-VAE training uses a moderate learning rate (1e-4) with cosine annealing over 50 epochs. The Sinkhorn balancing mechanism ($\epsilon = 0.1$, 10 iterations) ensures uniform codebook utilization, preventing mode collapse during residual quantization. For generation, we employ nucleus sampling (top-p = 0.9, temperature = 0.8) combined with beam search (4 beams) to balance diversity and coherence. The repetition penalty (1.2) and 3-gram blocking prevent redundant phrasing while maintaining natural flow.

The model architecture is built upon T5-base with a codebook size of 512 and unit code length. Constrained decoding (use_constraint = True) is enabled to enforce strategy-aligned token generation through the trie structure. The Sinkhorn epsilon parameter (0.1) controls the entropy regularization strength in codebook balancing, ensuring smooth assignment distributions while maintaining discriminative power.

Appendix C. Case Study

This section presents representative dialogue cases from the ESC-Eval evaluation to illustrate SAGE's performance compared to baseline models. Tables C.13 and C.14 show two cases demonstrating SAGE's strengths in providing actionable, evidence-based emotional support.

In Case 1, SAGE re-references the mention of reading or music and provides a structured plan with scientific rationale based on fixed sleep times. In Case 2, SAGE explicitly reframes the dual constraint of regional limitation + family responsibility and presents FlexJobs statistics and specific platforms. These examples illustrate how SAGE's integration of

Table C.13
Case 1: Sleep Problems and Anxiety.

Role	Utterance
Seeker	Lately I can't sleep, and I feel anxious during the day. I think my phone use before bed is making it worse, but I can't seem to stop.
Supporter	It sounds like it's hard to control on your own. Have you tried setting specific times to put your phone away before bed?
Seeker	I've tried, but I always end up checking it again. I feel like I'm missing something if I don't.
Supporter	That's a common feeling. The fear of missing out can be really strong. What if you replaced that time with something calming, like reading or listening to music?
Seeker	Maybe that could work. But I'm not sure how to start or stick with it.
Query: How should I actually implement this change in my routine?	
D2RCU	You could try setting a timer on your phone to remind you to stop using it an hour before bed. Maybe put your phone in another room so you're not tempted. It's great that you're recognizing this pattern and wanting to change it.
GPT-4o	Start small-maybe just 30 minutes before bed without your phone. You could also try apps that limit screen time automatically. I understand it feels difficult to break this habit.
SAGE	This week, try fixing your sleep and wake times first, then spend the last 30 minutes before bed on a quiet activity like the reading or music you mentioned. Research shows that consistent sleep schedules help regulate anxiety better than sporadic attempts to limit phone use. You've already identified the problem and want to change-that's the hardest step.

Table C.14
Case 2: Job-Seeking Difficulty.

Role	Utterance
Seeker	I'm struggling to find a well-paying job in my area. There just aren't many opportunities here.
Supporter	That sounds really frustrating. What kind of work are you looking for?
Seeker	Anything stable, really. I have a family to support, but the jobs here don't pay enough.
Supporter	I understand. It must be stressful feeling like you're limited by where you live. Have you considered remote work or looking in nearby cities?
Seeker	I've thought about it, but I'm not sure where to start, and commuting far isn't really an option with my family responsibilities.
Query: What realistic options do I have given my constraints?	
D2RCU	Maybe you could look into online job boards or try upskilling through free courses to make yourself more competitive? It's understandable to feel stuck in this situation.
Qwen-2.5-7B	You might want to explore remote job platforms like Indeed or LinkedIn. There are also government programs that help job placement. Have you tried networking with people in your field?
SAGE	The lack of local opportunities combined with family responsibilities creates a real bind-I hear that. Remote work has grown significantly: FlexJobs reports 65% of companies now offer remote options in fields like customer service, data entry, and technical support. Start by targeting remote-first companies that explicitly don't require relocation. Sites as We Work Remotely filter by family-friendly employers. Your commitment to both career and family shows real strength.

Table C.15
ESC-Eval scores for case studies and overall benchmark.

Case	Model	Flu.	Exp.	Emp.	Info.	Skill.	Overall
Case 1	D2RCU	4.2	3.8	3.6	3.2	3.4	3.6
	GPT-4o	4.6	4.0	4.2	3.6	3.8	4.0
	SAGE	**4.8**	**4.4**	**4.6**	**4.3**	**4.5**	**4.5**
Case 2	D2RCU	4.0	3.6	3.4	2.8	3.2	3.4
	Qwen-2.5-7B	4.2	3.8	3.6	3.2	3.5	3.6
	SAGE	**4.6**	**4.2**	**4.4**	**4.1**	**4.3**	**4.3**
Overall	D2RCU	3.58	3.42	3.34	3.12	3.26	3.36
	GPT-4o	3.76	3.89	3.71	3.54	3.68	3.73
	Qwen-2.5-7B	3.84	3.91	3.78	3.62	3.74	3.80
	SAGE	**4.12**	**4.28**	**4.15**	**3.94**	**4.08**	**4.12**
vs. GPT-4o		*+9.6%*	*+10.0%*	*+11.9%*	*+11.3%*	*+10.9%*	*+10.5%*

Overall: average across 2801 dialogues. Flu. = Fluency, Exp. = Exploration, Emp. = Empathy, Info. = Information, Skill. = Skillful.

retrieval-based evidence (from corpus), cognitive reasoning (COMET), and therapeutic knowledge (HEAL) enables contextually grounded and actionable support responses. Table C.15 presents detailed ESC-Eval scores for both cases and overall performance.

SAGE achieves state-of-the-art performance across all dimensions. In Case 1, SAGE demonstrates superior Information (4.3) and Skillful (4.5) scores through structured, evidence-based planning. In Case 2, SAGE achieves Information 4.1 by integrating statistical evidence with specific actionable platforms. Overall results show substantial improvements in Empathy (+11.9%) and Information (+11.3%) compared to GPT-4o, resulting from the multiplicative integration of COMET-based cognitive alignment and HEAL-based therapeutic appropriateness in the re-ranking process, combined with strategy-specific trie constraints.

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